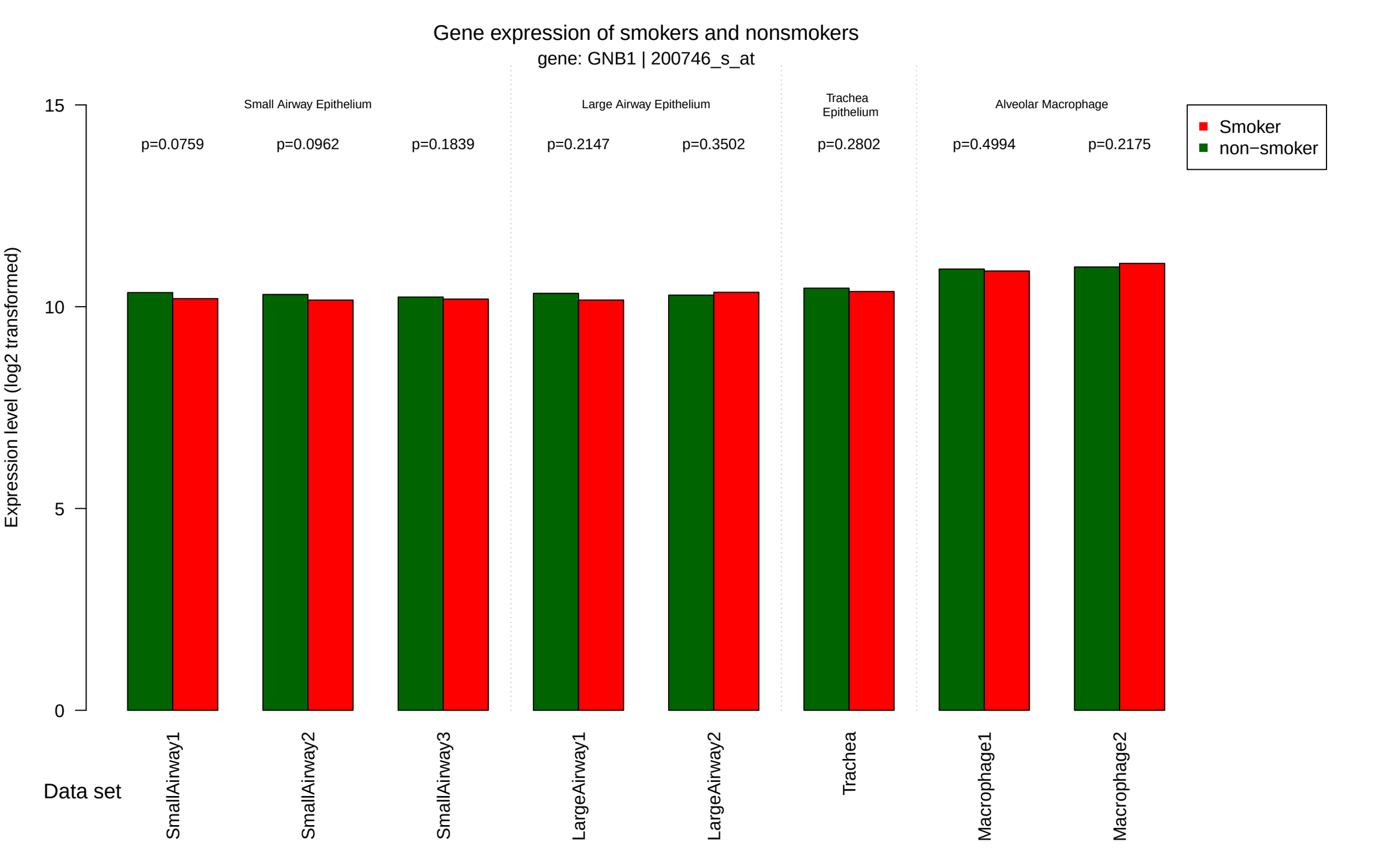
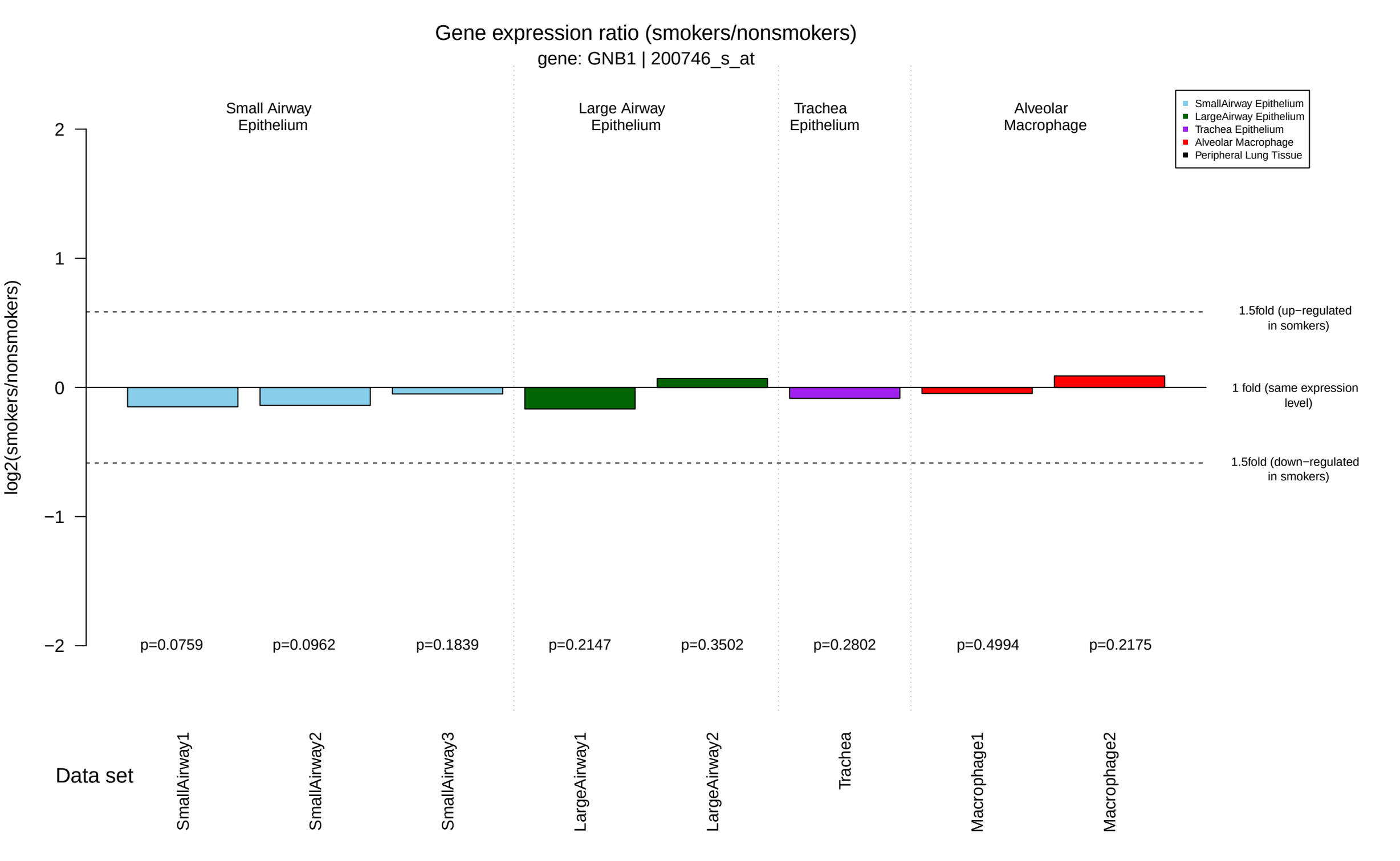
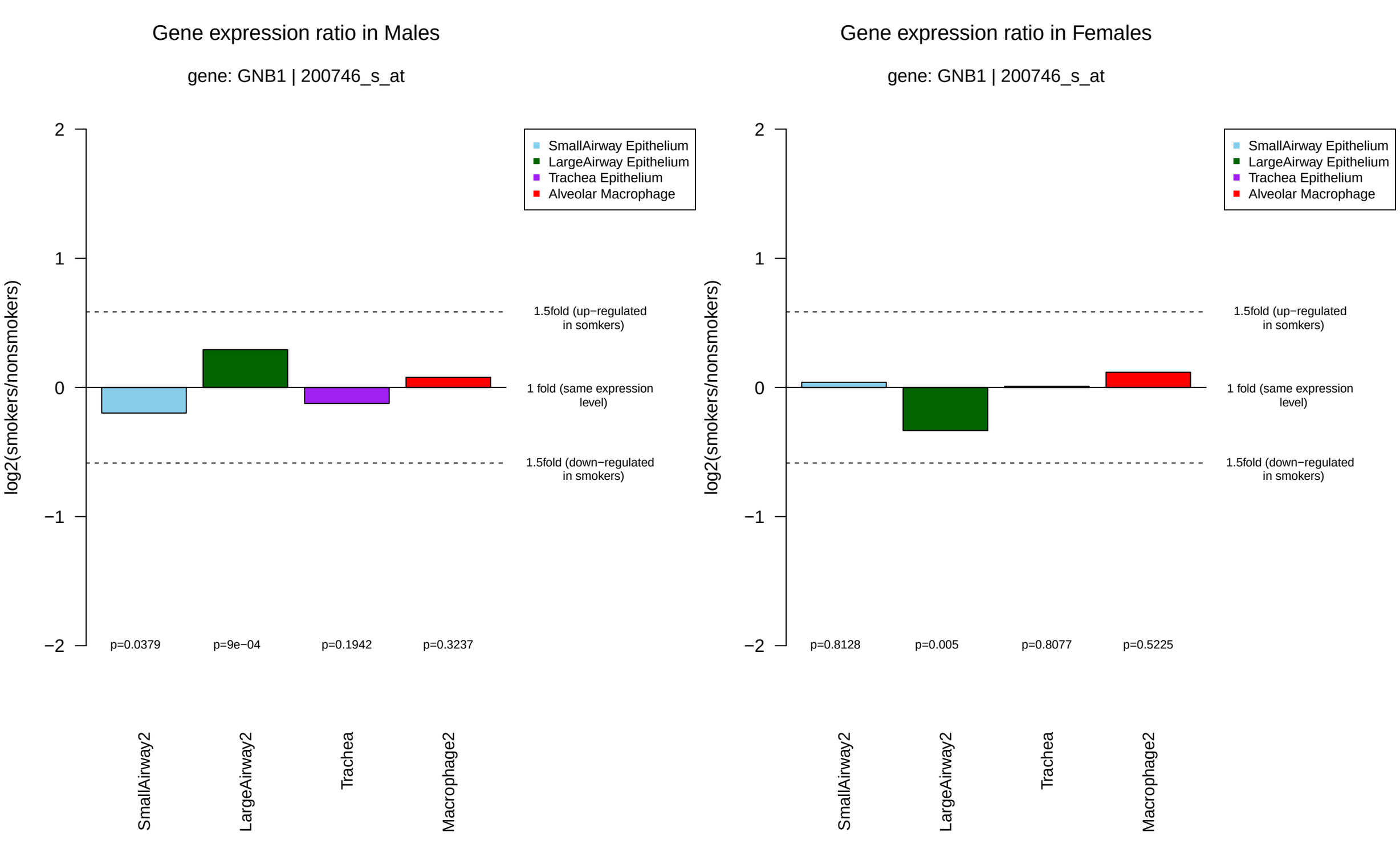
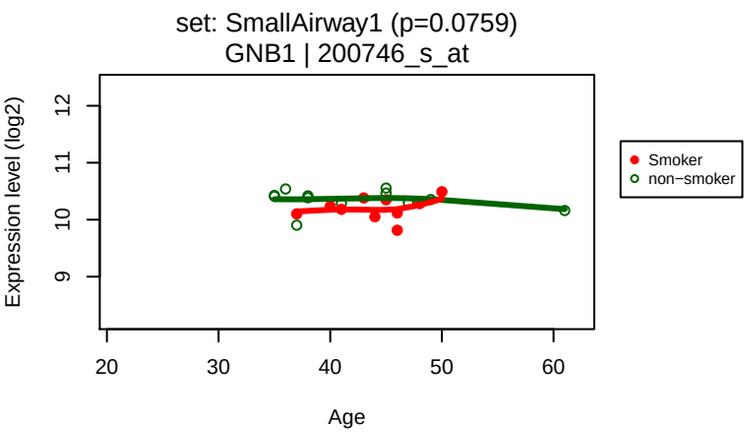
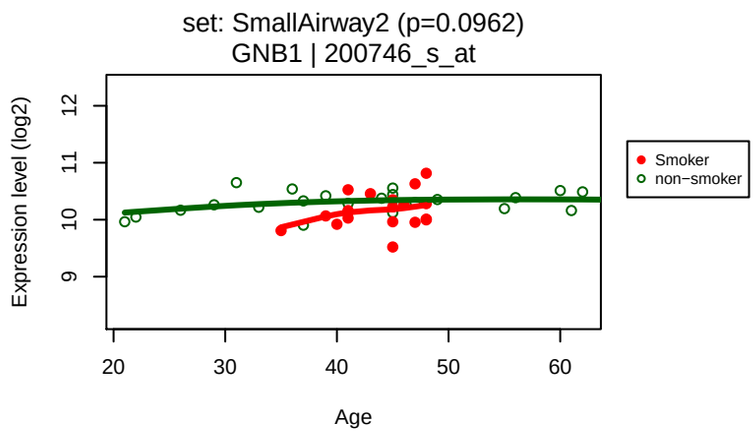
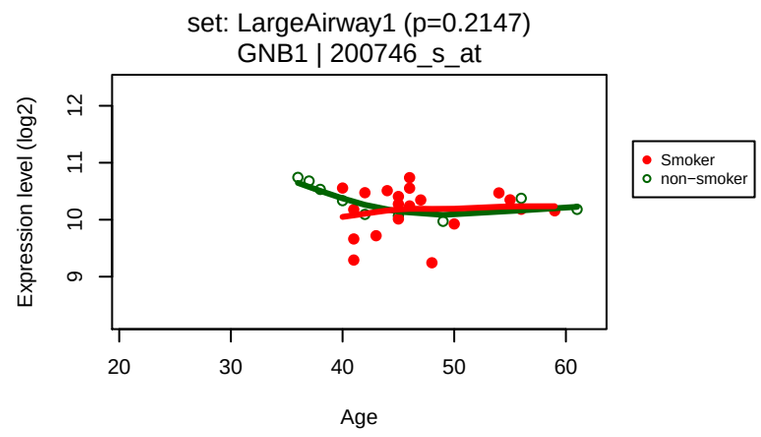

Gene expression of smokers and nonsmokers
gene: GNB1 | 200746_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0759
p=0.0962
p=0.1839
p=0.2147
p=0.3502
p=0.2802
p=0.4994
p=0.2175
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

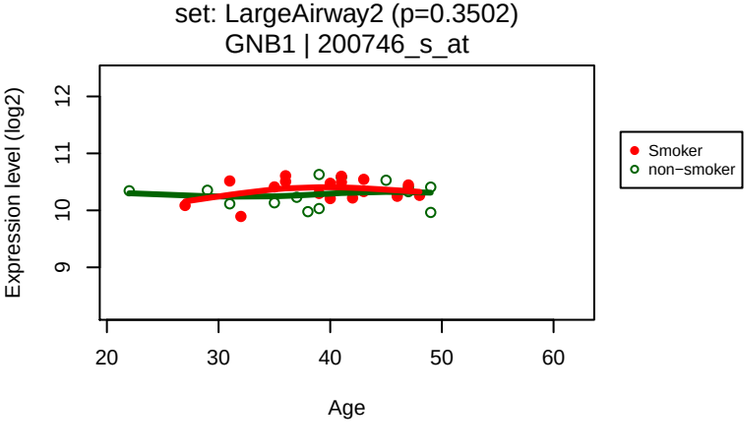

Gene expression ratio (smokers/nonsmokers)
gene: GNB1 | 200746_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0759
p=0.0962
p=0.1839
p=0.2147
p=0.3502
p=0.2802
p=0.4994
p=0.2175
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

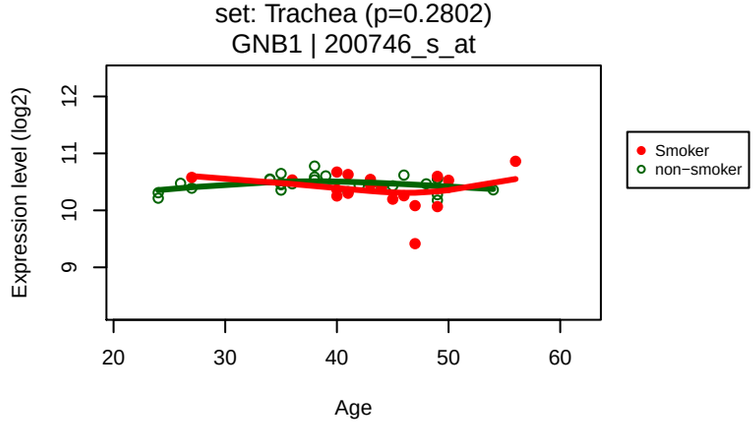

Gene expression ratio in Males
gene: GNB1 | 200746_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0379
p=9e−04
p=0.1942
p=0.3237
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNB1 | 200746_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8128
p=0.005
p=0.8077
p=0.5225
SmallAirway2
LargeAirway2
Trachea
Macrophage2

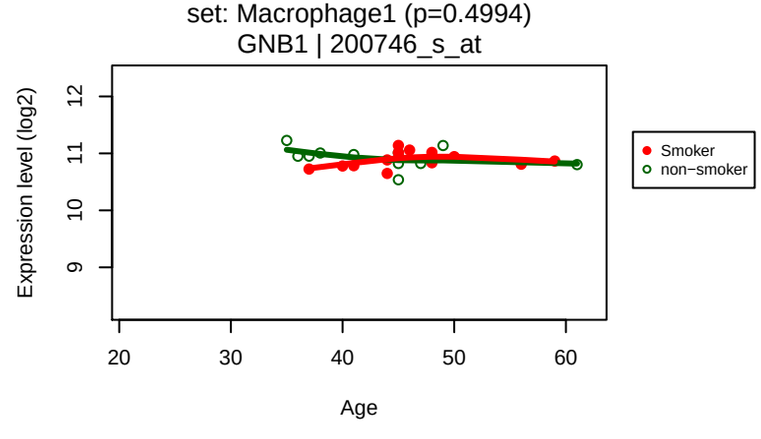

set: SmallAirway1 (p=0.0759)
GNB1 | 200746_s_at
Expression level (log2)
Age
Smoker
non−smoker

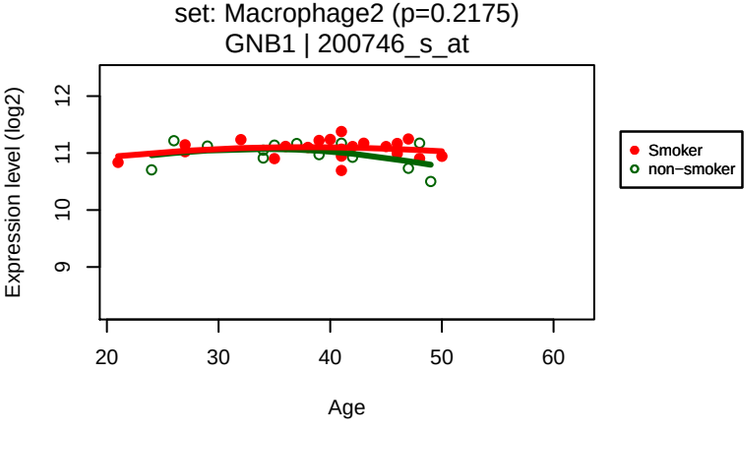

set: SmallAirway2 (p=0.0962)
GNB1 | 200746_s_at
Expression level (log2)
Age
Smoker
non−smoker

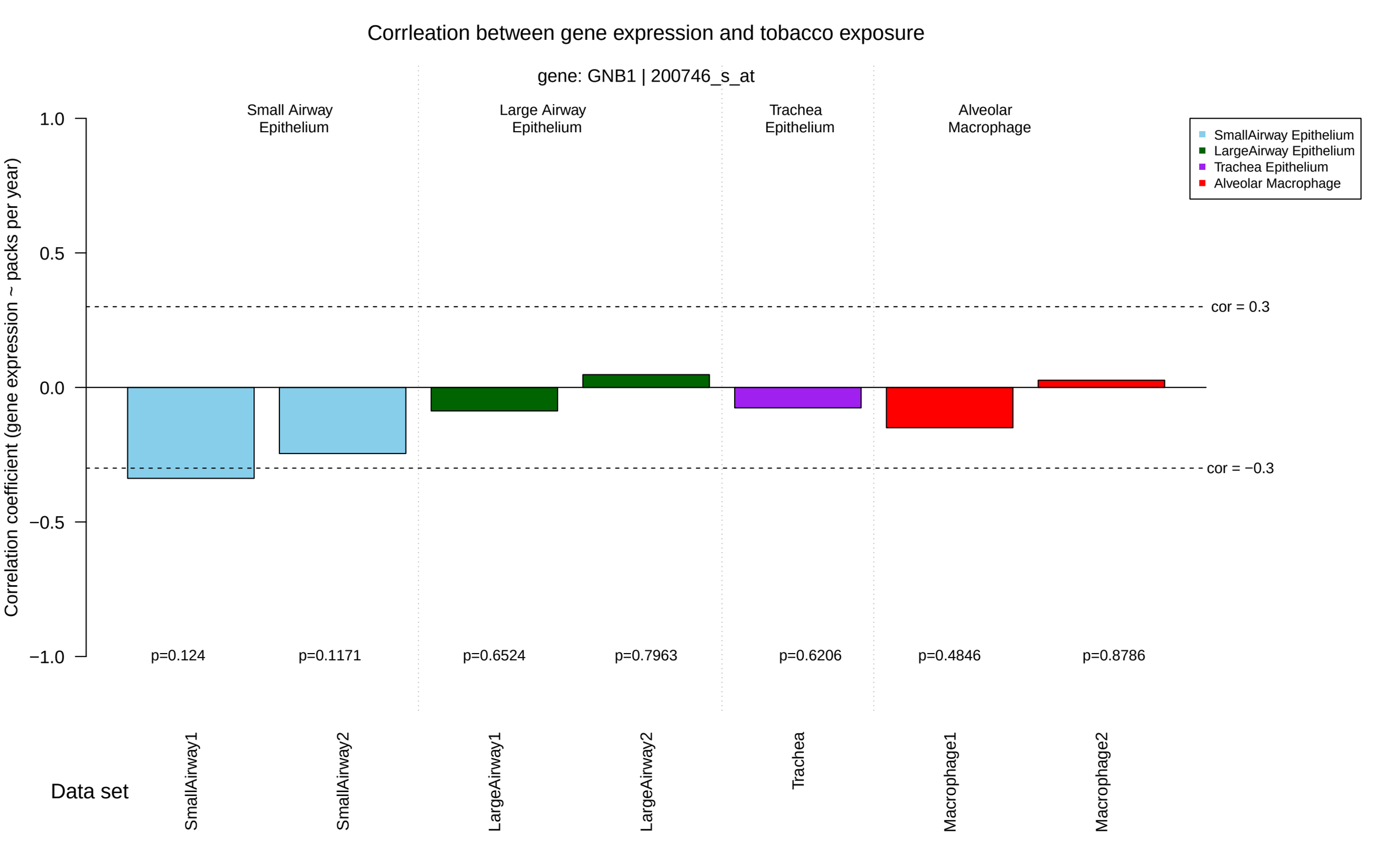

set: LargeAirway1 (p=0.2147)
GNB1 | 200746_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3502)
GNB1 | 200746_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2802)
GNB1 | 200746_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4994)
GNB1 | 200746_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2175)
GNB1 | 200746_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNB1 | 200746_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.124
p=0.1171
p=0.6524
p=0.7963
p=0.6206
p=0.4846
p=0.8786
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2